



«Approved»
by the Dean
Syzdykova Z.A.
» of _____ 2024

Syllabus
Academic Year 2024-2025

1. General Information	
Course title	Generative algorithms
Degree cycle (level)/major	7M06103 Applied data Analytics
Year, term	Year 2, term 4
Number of credits	4
Language of delivery:	English
Prerequisites	Methods and tools for data analysis / Introduction to neural networks / Data processing and understanding
Postrequisites	Reinforcement learning
Lecturer(s)	Aliya Shukirova, PhD in Automation and Control, Assistant Professor, a.shukirova@astanait.edu.kz, Astana IT University, Expo, C1 block, 2nd floor, office C1.1.337
2. Goals, Objectives and Learning Outcomes of the Course	
1. Course Description	“Generative algorithms” is a 10-week course designed to introduce students foundational principles and techniques that enable machines to generate complex outputs, such as text, images, and music. Students will explore the history and evolution of generative AI, including landmark models and milestones that have shaped the field.
2. Course Goal(s)	Modern data processing is not possible without parallel systems and really big problems require distributed computing systems to be involved. The goal of the course is to introduce the students into important basic concepts of parallel computing and provide practice to obtain better understanding of the problems. This creates a basis for the students to learn more advanced concepts and other technologies in more details in preparation either for a job in industry or for the following courses at the graduate level.
3. Course Objectives:	Students who take this course will be given the skills they need to comprehend cutting-edge generative models and develop their own by covering both the theoretical underpinnings and practical applications of generative AI.
4. Skills & Competences	• To learn the fundamentals of Generative algorithms and Deep Learning. • To explore Generative AI models like GANs, VAEs, and Transformers. • To discuss current trends and future directions in GenerativeAI research.

5. Course Learning Outcomes:	By the end of this course students will be able to: • Understand the fundamental ideas of generative AI, such as deep learning, probabilistic modelling, and generative models. • Learn about advanced training and assessment methods for generative models, such as optimisation strategies, loss functions, and evaluation metrics. • Learn about the ethical dilemmas and difficulties raised by generative AI, such as those with bias, justice, and privacy.
6. Methods of Assessment	The expected learning outcomes for the course will be assessed through graded activities and ungraded activities. The graded activities include exam and 6 practical assignments. The ungraded activities will be used to monitor your progress. A variety of these ungraded assessment techniques may be employed, including problems to be completed during class, direct questioning of students, answering students questions in class, one-minute classroom assessment techniques, and discussions during office hours.
7. Reading List	<u>Basic Literature:</u> 1. Foster D. Generative Deep Learning: Teaching Machines to Paint. Write, Compose, and Play O'Reilly Media Incorporated. 2019:139-40. 2. Goodfellow I, Bengio Y, Courville A. Deep learning. MIT press; 2016 Nov 10. 3. Foster D. Generative deep learning. "O'Reilly Media, Inc."; 2022 Jun 28. 4. Tomczak JM. Deep generative modeling. Springer; 2022. 5. Babcock J, Bali R. Generative AI with Python and TensorFlow 2: Create images, text, and music with VAEs, GANs, LSTMs, Transformer models. Packt Publishing Ltd; 2021 Apr 30. <u>Supplementary literature:</u> 1. Manaswi NK. Generative Adversarial Networks with Industrial Use Cases: Learning How to Build GAN Applications for Retail, Healthcare, Telecom, Media, Education, and HRTech. BPB Publications; 2020 Mar 4. 2. van der Zant T, Kouw M, Schomaker L. Generative artificial intelligence. Springer Berlin Heidelberg; 2013. 3. Ahirwar K. Generative adversarial networks projects: Build next-generation generative models using TensorFlow and Keras. Packt Publishing Ltd; 2019 Jan 31.
1. Resources	Online journals, article, papers, books and internet resources as well as online emulators and online software for simulation.
2. Course policy	Course and University policies include: Attendance: Attendance is not allocated any grading points in the marking scheme, but is compulsory to pass the course. Normally students are required to achieve course attendance of minimum 70% to get admitted to the examination rubric. In case a student misses 30% or more class sessions without a valid excuse the instructor has the right to mark him as "not graded". In such case a student is not admitted to the exam and automatically fails the course.

In cases, when a student misses class session due to valid reasons (is excused by the instructor or the dean's office) he or she has to confirm the absence reason using a valid document in accordance with the academic policy of AITU.

It should be NOTED that in cases when a student is excused for 30% of the scheduled class sessions or more he or she has to study material provided under the course on their own. Course instructor might provide additional opportunities to submit missed graded pieces of work during office hours or conduct alternative assessment exercises using method of his or her choosing.

Preparation for Class: Class participation is a very important part of the learning process in this course. Although not explicitly grade, students will be evaluated on the QUALITY of their contributions and insights. Quality comments possess one or more of the following properties:

- Offers a different and unique, but relevant, perspective;
- Contributes to moving the discussion and analysis forward;
- Builds on other comments.

Class work: The duration of each lecture and practical lesson is 50 minutes for offline class, and 40 minutes for online class. Students are expected to complete all readings and assignments ahead of time, attend class regularly and participate in class discussions. In case of systemic student's misconduct, the student can be dispensed from the classes.

Being late on class: When students come to class late, it can disrupt the flow of a lecture or discussion, distract other students, impede learning, and generally erode class morale. Moreover, if left unchecked, lateness can become chronic and spread throughout the class. Therefore, the being late to the class is not welcome and can have restriction activities by the course instructor.

Attestation I and II: Students, who score less than 25% for Attestation period I or Attestation period II (RK1/RK2) automatically fail the course.

Home work / Assignments: The assignments are designed to acquaint students with the theoretical knowledge and practical skills required for the course. The textbook readings will be supplemented with materials collected from recent professional articles and journals. In case of using someone's work (papers, articles, any publications), all works must be properly cited. Failure to cite work will be resulted as a cheating from the students and may be a subject of additional disciplinary measures.

Late submissions: Most assignments will be discussed in class on the due date. It is expected that all work will be submitted on time. All gradings are based using a percentage grading scale.

In the case of some extraordinary event, students should notify the course instructor and request an extension of the deadline for

	<i>submission. If approved, a new date will be given to the student depending upon the circumstances by the instructor.</i> Final exam* / Final project: The final exam for the course “Generative algorithms” will be taken in the form of an individual written exam. Laptops and mobile devices can only be used for classroom purposes when directed by the course instructor. Misuse of laptops or handheld devices will be considered a breach of discipline and appropriate action will be initiated by the instructor. Online lessons can be used in case if there won’t be a chance to make offline traditional lessons. It must not discourage the interest and enthusiasm of students. The main software to run the online lessons is Microsoft Teams for video calls and live webinars, and Moodle (moodle.astanait.edu.kz) as a Learning Management System. Also, some alternatives such as Telegram, Zoom, or other messenger may be involved as an additional workaround. Cheating and plagiarism are defined in the Academic conduct policies of the university and include: 1. Submitting work that is not your own papers, assignments, or exams; 2. Copying ideas, words, or graphics from a published or unpublished source without appropriate citation; 3. Submitting or using falsified data; 4. Submitting the same work for credit in two courses without prior consent of both instructors. Any student who is found cheating or plagiarizing on any work for this course will receive 0 (zero) for that work and further actions will also be taken regarding academic conduct policies of the university. Academic Conduct Policies of the university: The full texts of all the academic conduct code will be posted to the students using Learning Management System (moodle.astanait.edu.kz). Contacting the Course instructor: The easiest and most reliable way to get in touch with the course instructor is by email. Students must feel free to send email if you have a question related to the course. Instructor responds as soon as they can but not always instantaneously. Besides that, students are also welcomed to arrange a one-to-one meeting with the instructor by their office during office hours to discuss the class using both offline and online.
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* Final exam can be given by instructor in the any other forms such as Written Exam, Final Quiz, Oral Tasks and Exams, etc.

3. Course Content

#	Abbreviation	Meaning
1	ISIS	Instructor-supervised independent work
2	SIS	Students’ independent work
3	IP	Individual project

4	PA	Practical assignment
5	LW	Laboratory work
6	MCQ	Multiple choice quiz

3.1 Lecture, Practical/Seminar/Laboratory Session Plans

Week No	Course Topic	Lectures (H/W)	Practice sessions (H/W)	Lab. sessions (H/W)	TSIS (H/W)	SIS (H/W)
1	Introduction to Generative algorithms. Deep Learning	2	2			
2	Variational Autoencoders	2	2			
3	Generative Adversarial Networks	2	2			
4	Autoregressive Models	2	2			
5	Normalizing Flow Models	2	2			
6	Energy-Based Models	2	2			
7	Diffusion Models	2	2			
8	Transformers.	2	2			
9	Advanced GANs.	2	2			
10	Music Generation	2	2			
	Total hours: 40	20	20			

3.2 List of Assignments for Student Independent Study

№	Assignments (topics) for Independent study	Hours	Recommended literature and other sources (links)	Form of submission
1	2	3	4	5
1	Convolutional Neural Network (CNN)	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
2	Analysis of the Variational Autoencoder	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
3	Conditional GAN (CGAN)	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
4	PixelCNN	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources

				Electronic version
5	Other Normalizing Flow Models: GLOW, FFIJORD	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
6	Analysis of the Energy-Based Model	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
7	Analysis of the Diffusion Model	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
8	Other Transformers: T5, GPT-3 and GPT-4, ChatGPT	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
9	Other Important GANs: Self-Attention GAN (SAGAN), BigGAN, VQ-GAN, ViT VQ-GAN	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version
10	MuseGAN	9	Reading list, books, internet resources Electronic version	Reading list, books, internet resources Electronic version

4. Student Performance Evaluation System for the Course

Period	Assignments	Number of points	Total
1 st attestation	Assignments: assignment 1, assignment 2, assignment 3, Mid Term	60 20 20 20 40	100
2 nd attestation	Assignments: assignment 4, assignment 5, assignment 6, End Term	60 20 20 20 40	100
Final exam	Written Exam		100
Total	0,3 * 1st Att + 0,3 * 2nd Att + 0,4*Final		100

Achievement level as per course curriculum shall be assessed according to the evaluation chart adopted by the academic credit system.

Letter Grade	Numerical equivalent	Percentage	Grade according to the traditional system
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A	4,0	95-100	Excellent
A-	3,67	90-94	
B+	3,33	85-89	
B	3,0	80-84	Good
B-	2,67	75-79	
C+	2,33	70-74	
C	2,0	65-69	Satisfactory
C-	1,67	60-64	
D+	1,33	55-59	
D	1,0	50-54	Fail
FX	0	25-49	
F	0	0-24	

Based on the specific grade for each assignment, and the final grade, following criteria must be satisfied:

Grade	Criteria to be satisfied
90-100	- Work would be worthy of further dissemination under appropriate conditions - Mastery of advanced methods and techniques at a level beyond that explicitly taught - Ability to synthesize and employ in an original way idea from across the subject - Outstanding command of critical analysis and judgment
80-89	- Excellent range and depth of attainment of intended outcomes - Mastery of a wide range of methods and techniques - Evidence of study and originality of what has been taught - Able to display a command of critical analysis and judgement
70-79	- Attained all the intended learning outcomes for a unit - Able to use well a range of methods and techniques to come to conclusions - Able to employ critical analysis and judgement
60-69	- Some limitations in attainment of learning objectives, but has managed to grasp most of them - Able to use most of the methods and techniques taught - Evidence of study and comprehension of what has been taught but grasp insecure - Some grasp of the issues and concepts underlying the techniques and material taught, but weak and incomplete
50-59	- Attainment of only a minority of the learning outcomes - Able to demonstrate a clear but limited use of some of the basic methods and techniques taught - Weak and incomplete grasp of what has been taught - Deficient understanding of the issues and concepts underlying the techniques and material taught
25-49	- Attainment of nearly all the intended learning outcomes deficient - Lack of ability to use at all or the right methods and techniques taught - Inadequately and incoherently presented - Wholly deficient grasp of what has been taught - Lack of understanding of the issues and concepts underlying the techniques and material taught
0-24	No significant assessable material, absent or assessment missing a must pass component

5. Methodological Guidelines

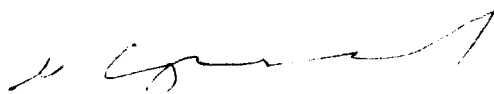
Assessment is administered continuously throughout the course. The students are rated against their performance in continuous rating administered throughout the semester (60%) and summative rating done during the examination session (40%), total 100%. Continuous rating is students' on-going performance in class and independent work. Class work is assessed for attendance, laboratory works' defense and in- class assessments.

- **ISIS (Instructor Supervised Student Independent Study)** -comprises presentation to be done by students independently and checked by instructor.
- **Mid-term and End-term** is a review of the topics covered and assessment of each student's knowledge. The form of the midterm and end term exams is complex.
- **Final assessment** is a combination of both individual (team) project (report) and oral presentation (slides) to evaluate the students' academic performance and professional skills. At the completion of this course each student has to submit an online project version conforming to the project outline, as well as to prepare and present a slide presentation that follows the presentation outline. Project iterations would be required to submit as well. Students should submit the written Report of Final project and slide-Presentation 3 days before the Final exam day.

6. Lecturer (lecturers) approvals Full name Job title Date Sign

Full name	Job title	Date	Sign
Aliya Shukirova	Assistant Professor	26/12/2024	

Director



Sergaziyev M.Zh.